

ASSIGNMENT NO. 10:

Window Functions and Analytical Queries

Questions

Q1) Create a MySQL database named SalesDB and a table named SalesData.

SalesData Table

Column Name	Data Type	Constraint
SaleID	INT	PRIMARY KEY
Salesperson	VARCHAR(50)	NOT NULL
Department	VARCHAR(30)	NOT NULL
SaleDate	DATE	NOT NULL
Amount	DECIMAL(10,2)	NOT NULL

Insert at least 12 sales records belonging to three different departments.

Query-

```
CREATE DATABASE SalesDB;
```

```
USE SalesDB;
```

```
CREATE TABLE SalesData (  
    SaleID INT PRIMARY KEY,  
    Salesperson VARCHAR(50) NOT NULL,  
    Department VARCHAR(30) NOT NULL,  
    SaleDate DATE NOT NULL,  
    Amount DECIMAL(10,2) NOT NULL  
);
```

```
INSERT INTO SalesData  
(SaleID, Salesperson, Department, SaleDate, Amount)  
VALUES  
(1, 'Rahul Sharma', 'Electronics', '2026-01-05', 75000.00),  
(2, 'Priya Patil', 'Electronics', '2026-01-10', 85000.00),
```

(3, 'Aman Verma', 'Electronics', '2026-01-15', 65000.00),
 (4, 'Sneha Joshi', 'Clothing', '2026-01-07', 55000.00),
 (5, 'Karan Mehta', 'Clothing', '2026-01-12', 70000.00),
 (6, 'Neha Shah', 'Clothing', '2026-01-18', 70000.00),
 (7, 'Rohan Desai', 'Furniture', '2026-01-06', 90000.00),
 (8, 'Meera Kulkarni', 'Furniture', '2026-01-11', 60000.00),
 (9, 'Arjun Nair', 'Furniture', '2026-01-20', 80000.00),
 (10, 'Divya More', 'Electronics', '2026-01-22', 95000.00),
 (11, 'Vikas Gupta', 'Clothing', '2026-01-25', 50000.00),
 (12, 'Pooja Singh', 'Furniture', '2026-01-28', 75000.00);

Q2) Write a MySQL query using ROW_NUMBER() to assign a unique row number to every salesperson.

The result should be arranged by sale amount from highest to lowest.

Required Output

Column
Salesperson
Department
Amount
SalesRowNumber

Query-

SELECT

Salesperson,

Department,

Amount,

ROW_NUMBER() OVER (ORDER BY Amount DESC) AS SalesRowNumber

FROM SalesData

ORDER BY Amount DESC;

Output-

Salesperson	Department	Amount	SalesRowNumber
Divya More	Electronics	95000.00	1
Rohan Desai	Furniture	90000.00	2
Priya Patil	Electronics	85000.00	3
Arjun Nair	Furniture	80000.00	4
Pooja Singh	Furniture	75000.00	5
Rahul Sharma	Electronics	75000.00	6
Karan Mehta	Clothing	70000.00	7
Neha Shah	Clothing	70000.00	8
Aman Verma	Electronics	65000.00	9
Meera Kulkarni	Furniture	60000.00	10
Sneha Joshi	Clothing	55000.00	11
Vikas Gupta	Clothing	50000.00	12

Q3) Write a MySQL query using RANK() to rank salespersons according to their sale amount.
Students having the same sale amount should receive the same rank.

Required Output

Column
Salesperson
Department
Amount
SalesRank

Query-

SELECT

Salesperson,

Department,

Amount,

RANK() OVER (ORDER BY Amount DESC) AS SalesRank

FROM SalesData

ORDER BY Amount DESC;

Output-

Server: 127.0.0.1 » Database: collegedb » Table: SalesData

Extra options

Salesperson	Department	Amount	SalesRank
Divya More	Electronics	95000.00	1
Rohan Desai	Furniture	90000.00	2
Priya Patil	Electronics	85000.00	3
Arjun Nair	Furniture	80000.00	4
Pooja Singh	Furniture	75000.00	5
Rahul Sharma	Electronics	75000.00	5
Karan Mehta	Clothing	70000.00	7
Neha Shah	Clothing	70000.00	7
Aman Verma	Electronics	65000.00	9
Meera Kulkarni	Furniture	60000.00	10
Sneha Joshi	Clothing	55000.00	11
Vikas Gupta	Clothing	50000.00	12

Console

```

Press Ctrl+Enter to execute query
> SELECT StudentName, Marks FROM StudentResult WHERE Marks > ( SELECT AVG(Marks) FROM StudentResult );
> SELECT StudentID, StudentName, CourseID, Marks FROM StudentResult WHERE CourseID = ( SELECT CourseID FROM Course WHERE...
> SELECT StudentName, Marks, (SELECT AVG(Marks) FROM StudentResult) AS AverageMarks FROM StudentResult;

```

Q4) Write a MySQL query to rank salespersons separately within each department.

Use:

Required Component
RANK()
OVER
PARTITION BY Department
ORDER BY Amount DESC

Query-

SELECT

Salesperson,

Department,

Amount,

RANK() OVER (

PARTITION BY Department

ORDER BY Amount DESC

) AS DepartmentRank

FROM SalesData

ORDER BY Department, DepartmentRank;

Output-

Server: 127.0.0.1 » Database: collegedb » Table: SalesData

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Extra options

Salesperson	Department	1	Amount	DepartmentRank	2
Karan Mehta	Clothing		70000.00		1
Neha Shah	Clothing		70000.00		1
Sneha Joshi	Clothing		55000.00		3
Vikas Gupta	Clothing		50000.00		4
Divya More	Electronics		95000.00		1
Priya Patil	Electronics		85000.00		2
Rahul Sharma	Electronics		75000.00		3
Aman Verma	Electronics		65000.00		4
Rohan Desai	Furniture		90000.00		1
Arjun Nair	Furniture		80000.00		2
Pooja Singh	Furniture		75000.00		3
Meera Kulkarni	Furniture		60000.00		4

Q5) Write a MySQL query to calculate the running total of sales according to the sale date.

Required Output

Column
SaleID
Salesperson
SaleDate
Amount
RunningTotal

Query-

SELECT

SaleID,

Salesperson,

SaleDate,

Amount,

SUM(Amount) OVER (

ORDER BY SaleDate

) AS RunningTotal

FROM SalesData

ORDER BY SaleDate;

Output-

SaleID	Salesperson	SaleDate	Amount	RunningTotal
1	Rahul Sharma	2026-01-05	75000.00	75000.00
7	Rohan Desai	2026-01-06	90000.00	165000.00
4	Sneha Joshi	2026-01-07	55000.00	220000.00
2	Priya Patil	2026-01-10	85000.00	305000.00
8	Meera Kulkarni	2026-01-11	60000.00	365000.00
5	Karan Mehta	2026-01-12	70000.00	435000.00
3	Aman Verma	2026-01-15	65000.00	500000.00
6	Neha Shah	2026-01-18	70000.00	570000.00
9	Arjun Nair	2026-01-20	80000.00	650000.00
10	Divya More	2026-01-22	95000.00	745000.00

Q6) Write a MySQL query to display the highest-performing salesperson from each department.

The query should:

Requirement
Use ROW_NUMBER()
Partition records by department
Arrange sale amounts in descending order
Use a subquery
Display only the first-ranked salesperson from every department

Query-

SELECT

Salesperson,

Department,

Amount

FROM (

SELECT

Salesperson,

Department,

Amount,

ROW_NUMBER() OVER (

PARTITION BY Department

ORDER BY Amount DESC

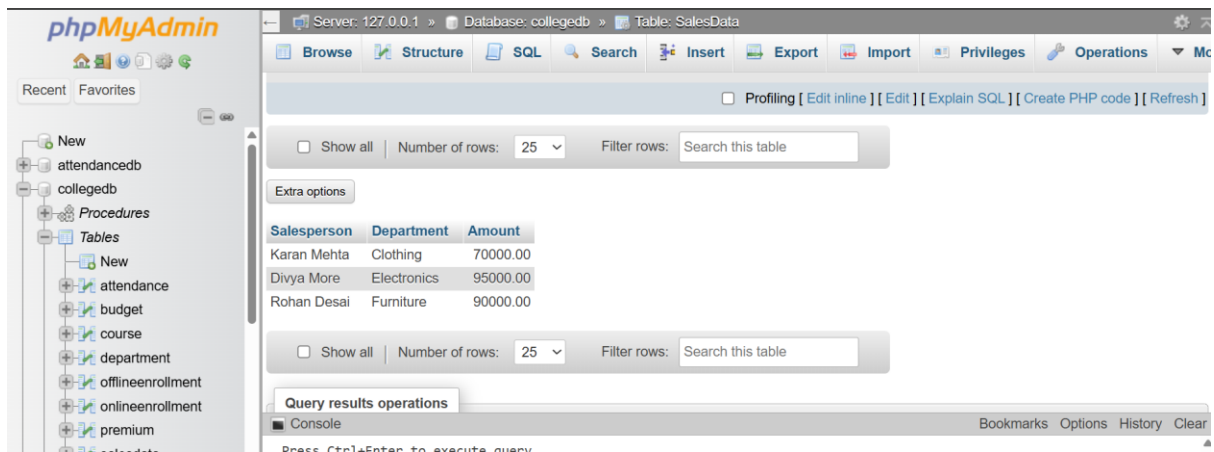
) AS RowNum

FROM SalesData

) AS RankedSales

WHERE RowNum = 1;

Output-



Server: 127.0.0.1 » Database: collegedb » Table: SalesData

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Salesperson	Department	Amount
Karan Mehta	Clothing	70000.00
Divya More	Electronics	95000.00
Rohan Desai	Furniture	90000.00

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Console Bookmarks Options History Clear

Press Ctrl+Enter to execute query